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Part 1a.

File locations:

- part1a/pcap1.pcap
- part1a/pcap2.pcap
- part1a/pcap3.pcap
- part1a/pcap4.pcap
- part1a/pcap5.pcap
- part1a/pcap6.pcap

1. List the different application layer protocols and their counts for each activity. In your report, specify how you figured out the protocol for each activity.

- 1: ICMP (40)
- 2: DNS (14), HTTPS (3208)
- 3: DNS (6), HTTP (66), QUIC (373)
- 4: HTTPS (8089), QUIC (2154)
- 5: DNS (6), FTP (10)
- 6: UDP (85)

2. How many HTTP and HTTPS packets did you record while performing activities 2 and 3?

2: 3208

3: 66

3. List the destination IP address used in each activity along with their timestamps. The destination IP address should be in the IPv4 format like x.x.x.x (e.g., “192.168.1.1”, “8.8.8.8”, “10.0.1.150”, etc.).

1: {'20.44.10.123', '13.66.138.105', '172.217.217.188', '140.82.116.6', '10.0.0.4'}

2: {'10.0.0.4', '172.217.217.188'}

3: {}

4: {'98.82.157.137', '104.18.26.193', '104.17.208.58', '151.101.43.52', '13.249.70.91', '23.206.229.111', '13.249.78.63', '10.0.0.4', '50.18.3.174', '52.45.101.96', '204.237.133.116', '44.244.162.28', '54.187.18.189', '34.211.131.132', '104.254.151.60', '18.173.121.78', '69.194.240.13', '151.101.0.134', '63.140.37.103', '44.239.221.55', '142.251.214.142', '54.187.74.22', '151.101.43.42', '13.249.74.121', '104.18.28.101', '3.169.183.5', '104.18.40.226', '35.186.253.211', '13.52.40.43', '52.33.195.54', '65.8.161.105'}

5: {'20.44.10.123', '13.66.138.105', '10.0.0.4', '17.248.192.1'}

6: {}

4. For activities 2, 3, and 4, can you tell which browser was used for these activities from the captured packets?

No, because 2 is HTTPS connection so you won’t be able to see headers, for 3 no http requests were being made so I could not see, and 4 is not done through browser.

Part 1b.

File locations:

- part1b/analyze.py

Secret Information: 'my-secret': 'Zubair Rocks!!'

Both PCAP1_2 and PCAP1_3 primarily capture local network activity involving Spotify Connect and AirPlay service discovery between nearby devices such as an iPhone and a Roku TV. Numerous mDNS packets advertise services like _spotify-connect._tcp.local, _airplay._tcp.local, and _apple-mobdev2._tcp.local, showing that devices were broadcasting their availability for media streaming and pairing on the same wireless network. In PCAP1_2, this is the main activity, suggesting someone was using or had the Spotify app open while the devices exchanged discovery messages. In PCAP1_3, that same background Spotify and AirPlay traffic is present, but the capture also includes DNS lookups for google.com, a short TLS handshake, and a sequence of ICMPv6 echo and time-exceeded messages. These indicate that, alongside the ongoing Spotify discovery, the user performed a traceroute or ping test to Google to check IPv6 connectivity.